KLH UNIVERSITY HYDERABAD

Credit Card Fraud Detection System Using Logistic Regression, SVM And Isolation Forest

Department of Computer Science and Engineering
Machine Learning Project

Team Number: T.N. – 18

Guide Name: Mrs. Jyothsna Datti

Project Title : Credit Card Fraud Detection System Using Logistic Regression, SVM And Isolation Forest

INTRODUCTION

- Credit card fraud is a major challenge in modern financial systems.
- It can cause significant financial losses and reduce customer trust.
- This project uses machine learning techniques to identify and classify fraudulent transactions.

PROBLEM STATEMENT & OBJECTIVE

Problem: Credit card transactions generate large amounts of data, making manual identification of fraudulent transactions difficult.

Objective: Compare Logistic Regression, SVM and Isolation Forest for credit-card fraud detection.

METHODOLOGY

1. Preprocess the data (cleaning, normalization).
2. Split into training and testing sets.
3. Implement Logistic Regression, SVM and Isolation Forest.
4. Handle class imbalance using SMOTE / balanced subset.
5. Evaluate using Accuracy, Precision, Recall, F1-score and ROC-AUC.

DATASET

- **Original dataset:** 20,000 transactions, 26 attributes.
- **Balanced subset** (after under sampling): 678 records (339 legitimate + 339 fraudulent).
- Contains numerical and categorical transaction-related attributes.
- **Target variable:** fraud (binary classification).

MODEL COMPARISON

Metric	Logistic Regression	SVM	Isolation Forest
Accuracy	81.17%	80.38%	80.02%
Precision	10.62%	11.05%	34.38%
Recall	86.83%	83.25%	17.75%
F1-score	18.07%	19.41%	23.2%
AUC-ROC	92.40%	92.63%	94.12%

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RESULTS – FRONT END

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1. Detection Form



2. Prediction Result



3. Model Performance

Logistic Regression		SVM		Isolation Forest	
Accuracy	88.17%	Accuracy	80.38%	Accuracy	80.02%
Precision	10.62%	Precision	11.05%	Precision	34.38%
Recall	86.83%	Recall	83.25%	Recall	17.75%
F1 Score	18.07%	F1 Score	19.41%	F1 Score	23.3%
AUC-ROC	92.40%	AUC-ROC	92.63%	AUC-ROC	94.12%

4. System Summary

- **25** input attributes
- **3** ML models (Logistic Regression, SVM, Isolation Forest)
- **2** outcomes: SAFE or FRAUD

5. Key Features

- Multiple ML models
- Transaction-risk inputs
- SAFE / FRAUD output
- Performance comparison
- Responsive interface

Conclusion The system compares supervised classification and anomaly detection to support automated identification of suspicious transactions.

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